Anahita Alimoradpour

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Subject: Application for PhD position - Bayesian Modeling and Counterfactual Methods for Nanoparticle Transport in Human Tissue (SYNERGIE project / ID Job: 3924)

Dear Dr. Azimi and the Selection Committee,

Computational transport models often treat nanoparticles as static entities moving through rigid biological systems. Yet, in biological environments, nanoparticles can change: they adsorb serum proteins, form dynamic coronas, and aggregate, altering their transport and tissue distribution. Models based on fixed diffusion parameters or deterministic ODEs may produce mathematically neat predictions but fail to capture tissue deposition accurately. This challenge drew me to the SYNERGIE project, which aims to develop computational frameworks that account for changes in nanoparticle identity across physiological barriers, a key step toward improving clinical translation.

My background bridges molecular mechanisms and quantitative modeling. I completed my M.Sc. in Medical Biotechnology at Tehran University of Medical Sciences as the top graduate (GPA: 19.93/20.00), building reproducible Python workflows with PyTorch and RDKit to model nonlinear biomolecular properties. Additionally, my B.Sc. training in clinical laboratory science provided a strong foundation in human pathology and cellular diagnostics. This perspective ensures I view physiological barriers not as abstract boundary conditions, but as dynamic systems shaped by biological variability and clearance.

This background has prepared me to contribute to the project's computational challenges. Nanoparticle transport is an underdetermined problem, with clinical and animal data that are often sparse and noisy. Bayesian inference can help quantify and propagate uncertainty rather than conceal it in point estimates, while counterfactual methods can examine how changes in surface charge, corona evolution, or tissue perfusion may affect delivery. My research has emphasized rigorous statistical validation, and I hope to bring that perspective to the computational work of SYNERGIE.

Joining Dr. Sepinoud Azimi's group at TU Delft offers an opportunity to deploy my wet-lab literacy and machine learning experience within a high-impact Horizon Europe consortium. I am ready to commit fully to this project and would welcome the chance to discuss my background and technical approach with you.

Sincerely,

Anahita Alimoradpour